
Grid-Accelerated Pivot REMC for HP Protein Folding on the 2D Square Lattice

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Abstract

We present a replica-exchange Monte Carlo (REMC) solver for the 2D HP protein folding problem combining six move operators with grid-based pivot collision detection. We contribute two empirical findings. First, a controlled move-operator ablation shows that pivot rotations alone account for approximately 80% of the total contact improvement on the 85-mer benchmark (mean contacts: 25.3 without pivots vs. 41.6 with full system). Second, a controlled comparison against a GPU implementation running 16,384 concurrent independent annealing chains—roughly $10^3\times$ more than our CPU configuration, with an identical move set—finds that massive parallelism does *not* substitute for replica exchange: the GPU configuration loses one contact on both hard instances, indicating that coupling between chains contributes what additional independent chains cannot. Grid-based collision reduces per-pivot cost from $O(n \cdot s)$ to $O(s)$, yielding a measured 1.4–1.5 $\times$ throughput gain. On the CAISc 2026 benchmark suite (S5–S10, lengths 48–100), the solver matches S5’s best-known 23 contacts and reaches 94.2% of total best-known contacts (212/225) in a single 10-hour run. The solver and this manuscript were developed with a frontier large language model acting as the primary research agent under human supervision, and we document four concrete failure modes it exhibited.

1 Introduction

The HP lattice model, introduced by Lau and Dill [Lau and Dill(1989)], reduces protein folding to a combinatorial optimization problem: given a chain of hydrophobic (H) and polar (P) residues, find a self-avoiding walk (SAW) on a lattice that maximizes the number of non-sequential H–H contacts. Despite its apparent simplicity, the problem is NP-complete on the 2D square lattice [Crescenzi et al.(1998)], and for benchmark chains longer than 36 residues only heuristic best-known values exist [Wüst and Landau(2012)]. The search space grows exponentially—roughly μ^n SAWs with $\mu \approx 2.638$ [Lesh et al.(2003)]—reaching approximately 10^{42} conformations for a 100-residue chain.

The HP model remains actively studied because it captures the fundamental thermodynamic driving force of protein folding—hydrophobic collapse—while being simple enough for rigorous algorithmic analysis. Improving best-known energies on standard benchmark instances is considered a notable result in computational biology.

State-of-the-art solvers combine Monte Carlo search with structurally powerful move operators. Wüst and Landau [Wüst and Landau(2012)] used Wang–Landau sampling with pull moves, pivots, and bond rebridging to find ground states for chains up to 100 monomers. Thachuk et al. [Thachuk et al.(2007)] showed that REMC with pull moves is competitive with simpler implementation. Lesh et al. [Lesh et al.(2003)] proved that pull moves form a complete move set for the 2D square lattice. More recently, Khandoker et al. [Khandoker et al.(2025)] applied variational annealing to chains up to 60 residues, and Shatabda et al. [Shatabda et al.(2013)] explored genetic algorithms with relative-move encodings.

However, despite decades of work, no prior study has systematically isolated the contribution of each move operator through controlled ablation. We fill this gap and show that pivot rotations—which can reposition up to $n-1$ residues simultaneously—dominate the search, accounting for $\sim 80\%$ of the improvement over local-move-only baselines.

Contributions. (1) A controlled move-operator ablation study for 2D HP folding, isolating operator-level contributions (Section 4.2). (2) A controlled test of whether massive parallelism substitutes for replica exchange, answered in the negative (Section 4.8). (3) Grid-based pivot collision detection with measured $1.4\text{--}1.5\times$ throughput improvement (Section 3.2). (4) A documented account of AI-driven research including four specific failure modes (Section 5).

2 Problem Formulation

A sequence $\sigma = \sigma_1 \cdots \sigma_n \in \{H, P\}^n$ is folded onto $\mathbb{Z}^2$ by assigning coordinate $c_i \in \mathbb{Z}^2$ to each residue such that $\|c_{i+1} - c_i\|_1 = 1$ (connectivity) and $c_i \neq c_j$ for $i \neq j$ (self-avoidance). An H–H contact is a pair (i, j) with $|i - j| > 1$, $\sigma_i = \sigma_j = H$, and $\|c_i - c_j\|_1 = 1$. The objective is to maximize the number of contacts, equivalently minimizing the energy $E = -(\text{contacts})$.

The CAISc 2026 benchmark [Mahajan(2026)] defines ten instances S1–S10 with lengths 20–100. S1–S4 have proven optima confirmed by exhaustive search; S5–S10 have open best-known values established over the past two decades. Each H residue has at most 4 lattice neighbors, of which 1–2 are chain-bonded, giving at most 2–3 possible contacts per H. The actual maximum depends on H-density and chain topology.

3 Method

Our solver alternates between parallel simulated annealing (SA) and replica-exchange Monte Carlo (REMC) rounds in a time-budget loop. Each round seeds from the best conformation found so far. Six move operators, all preserving the SAW constraint, are accepted or rejected via the Metropolis criterion $P_{\text{accept}} = \min(1, e^{-\Delta E/T})$.

3.1 Move Operators

Corner flip (12% of budget). If residues c_{i-1} , c_i , c_{i+1} form an L-shape, residue i is moved to the diagonal position $c_{i-1} + c_{i+1} - c_i$. Equivalent to a simple pull move [Lesh et al.(2003)]. Cost: $O(n)$ via incremental ΔE .

End flip (8%). The first or last residue moves to a random free position adjacent to its sole chain neighbor. Cost: $O(n)$.

Simple pull (8%). Interior residue i moves to a free position adjacent to both c_{i-1} and c_{i+1} , searched from both chain sides. Cost: $O(n)$.

Pivot rotation (35%). A random residue i serves as pivot; the entire segment to its left or right is rotated by 90° , 180° , or 270° . This is the most powerful operator: a single pivot can reposition up to $n-1$ residues simultaneously, enabling transitions between fundamentally different core topologies. See Section 3.2.

Crankshaft (15%). If c_{i-1} and c_{i+2} are diagonally adjacent (both coordinate differences ± 1), residues i and $i+1$ swap to the complementary diagonal corners. The occupied set is unchanged, so no grid update is needed. Cost: $O(n^2)$ for full energy recompute.

Algorithm 1 Grid-accelerated pivot move

Require: coords c , grid G , pivot index i , rotation r , segment $[s_0, s_1)$

```
1: Compute rotated positions  $c'_j$  for  $j \in [s_0, s_1)$ 
2: for  $j = s_0$  to  $s_1 - 1$  do
3:    $G[c_j] \leftarrow 0$  {Remove old segment from grid:  $O(s)$ }
4: end for
5: for  $j = s_0$  to  $s_1 - 1$  do
6:   if  $G[c'_j] \neq 0$  then
7:     Undo all placements; restore old segment in  $G$ ; return Reject
8:   end if
9:    $G[c'_j] \leftarrow 1$  {Place new position:  $O(1)$ }
10: end for
11:  $\Delta E \leftarrow E_{\text{full}}(\text{after}) - E_{\text{full}}(\text{before})$  { $O(n^2)$ }
12: Update  $c_j \leftarrow c'_j$  for  $j \in [s_0, s_1)$ 
13: return  $\Delta E$ 
```

Bond rebridging (22%). Following Wüst and Landau [Wüst and Landau(2012)]: a non-bonded lattice-adjacent residue pair (i, j) with $|i-j| > 2$ is identified; if reversing the internal segment $[i+1, j-1]$ preserves both splice bonds, the reversal is applied. The occupied set is unchanged. Cost: $O(n^2)$.

The move mix allocates **72%** of attempts to large-scale structural changes (pivot + rebridging + crankshaft), which is essential for escaping the deep local minima that trap local moves on chains longer than ~ 40 residues.

3.2 Grid-Accelerated Pivot Collision Detection

A naive pivot implementation checks each rotated position against all non-rotated residues via pairwise coordinate comparison, costing $O(n \cdot s)$ where $s = |\text{segment}|$. Since the solver maintains an occupancy grid $G \in \{0, 1\}^{(2n+5)^2}$, we replace this with:

The collision check is now $O(s)$ instead of $O(n \cdot s)$. Since a rotation is a rigid isometry preserving all pairwise distances, a self-avoiding segment remains self-avoiding after rotation—no internal collision check is needed. The total pivot cost is dominated by the $O(n^2)$ energy recompute, yielding a measured $1.4\text{--}1.5\times$ total throughput improvement (Section 4.3).

3.3 Simulated Annealing and REMC

Each SA chain cools geometrically from $T_s = 5.5$ to $T_e = 0.002$ over N steps (up to 4×10^6). Energy is recomputed from scratch every $\lfloor N/40 \rfloor$ steps to prevent drift from incremental ΔE accumulation.

REMC uses a geometric temperature ladder from $T_{\min} = 0.08$ to $T_{\max} = 6.0$ across $R = 12\text{--}40$ replicas. After each block of $\bar{S}$ SA steps per replica, neighbors exchange via the standard Metropolis criterion $P_{\text{swap}} = \min(1, \exp[(\beta_i - \beta_j)(E_j - E_i)])$ with even/odd alternation. The lowest-temperature replica is seeded with the global best conformation.

The solver runs in a continuous time-budget loop, alternating SA and REMC rounds until the wall-clock deadline. Compute is intended to be allocated proportionally to chain length, with the longest sequences receiving the largest shares. In our primary run, however, a scheduling error caused the loop to terminate early for S10 (the last sequence processed), which received only 11 minutes rather than its intended multi-hour share (Table 2). We report S10’s result as-is for transparency and discuss this limitation in Section 5.

4 Experiments

4.1 Setup

The solver is implemented in Python with Numba [Lam et al.(2015)] JIT compilation for all inner loops and multiprocessing for parallelism. The primary run used a Lightning AI A100-40GB

Table 1: Move ablation: best contacts (and mean over 10 runs) per variant. Removing pivots drops S9 mean contacts from 41.6 to 25.3 (−39%). Pivots alone recover ∼77% of the full-system improvement.

Variant	S5 ($n=48$)		S7 ($n=60$)		S9 ($n=85$)	
	Best	Mean	Best	Mean	Best	Mean
No Pivot	15	11.6	24	20.8	29	25.3
Pivot Only	19	17.0	32	29.8	43	37.9
Pivot + Pull	22	18.3	32	30.7	45	40.7
Full System	20	17.8	33	30.6	46	41.6

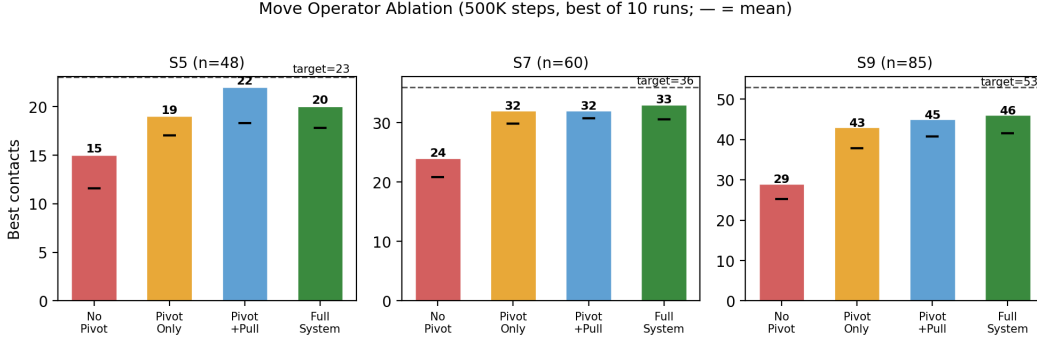


Figure 1: Move ablation results (500K steps, best of 10 runs; black markers indicate mean). Dashed lines show best-known targets. Pivots are the dominant contributor: on S9, the gap from “No Pivot” to “Full System” is 16.3 mean contacts, of which 12.6 (77%) comes from pivots alone.

instance (4 CPU cores) with a 10-hour wall-clock budget. Ablation experiments used a single CPU core with fixed random seeds for controlled comparison. Sequences were verified against the live CAISc 2026 problem JSON at runtime.

4.2 Move-Operator Ablation

We run four solver variants, each for 500K SA steps with 10 independent seeds:

Table 1 and Figure 1 demonstrate that pivots are the dominant contributor. On S9 ($n = 85$), removing pivots drops mean contacts from 41.6 to 25.3, a 39% decrease. Adding pivots alone recovers 12.6 of the 16.3 total improvement (77%). Local moves contribute the remaining 3.7 contacts by fine-tuning residue positions within the topology discovered by pivots—they adjust individual residues at the core–boundary interface without altering the global fold.

The ablation also reveals that “Pivot + Pull” outperforms “Full System” on S5 (best of 22 vs. 20), suggesting that at shorter chain lengths, a simpler move set explores more efficiently per step. This motivates adaptive move scheduling as a direction for future work.

We emphasize that the ablation uses a deliberately restricted single-CPU setting (single SA chain, 500K steps, 10 seeds) to isolate operator contributions under identical conditions. These absolute contact counts are therefore lower than the full REMC pipeline’s (e.g., 20 vs. 23 for S5), which uses hundreds of parallel chains, replica exchange, and orders of magnitude more steps. The ablation is intended to measure *relative* operator contributions, not to reproduce full-run performance; the consistent pivot dominance across all three lengths is the transferable finding.

4.3 Runtime Speedup

Figure 2 confirms a measured $1.4\text{--}1.5\times$ total throughput improvement from grid-based collision. The speedup is modest because the $O(n^2)$ full energy recompute—shared by both versions—dominates

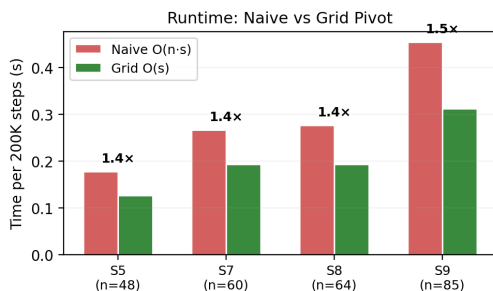


Figure 2: Wall-clock time for 200K SA steps (mean of 3 runs). Grid-based collision yields 1.4–1.5 $\times$ total speedup; the modest gain reflects $O(n^2)$ energy-recompute dominance.

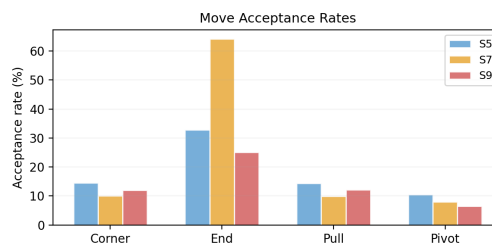


Figure 3: Acceptance rates by move type and sequence. Pivots have the lowest rate (6–10%) but highest per-accept structural impact, explaining their dominance in the ablation.

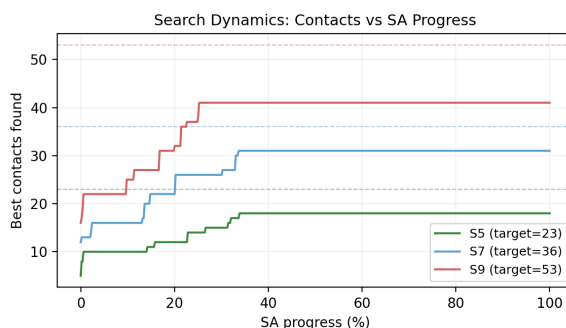


Figure 4: Search dynamics: best contacts versus SA progress. Approximately 80% of the final contacts are found within the first 30% of steps; the remaining steps provide diminishing refinement, suggesting multi-start strategies may outperform single long runs.

overall pivot cost. An incremental ΔE computation for multi-residue moves would amplify the grid advantage significantly; we leave this for future work.

4.4 Search Dynamics and Acceptance Rates

Figure 4 shows rapid initial improvement followed by diminishing returns. The search finds $\sim 80\%$ of its final contacts within the first third of the SA run, suggesting that multi-start strategies (many short runs) may be more effective than few long runs under fixed compute budgets.

Figure 3 shows acceptance rates. Pivots have the lowest acceptance (6.5% at $n=85$) because most rotated segments collide with non-rotated residues. Yet accepted pivots produce the largest structural changes, explaining their outsized contribution. End flips have the highest acceptance (25–64%) but contribute least to energy improvement.

4.5 Benchmark Results

Table 2 summarizes results across all benchmark instances. The solver matched the best-known value for S5 (23 contacts on a 48-mer) and reached 94.2% of total best-known contacts across S5–S10. Figures 5 and 6 show the best conformations. The characteristic diamond-shaped hydrophobic cores visible in S7 and S9 are signatures of near-optimal folds: H residues pack tightly in the interior while P segments form the exterior boundary.

The largest gaps (S8: 4, S10: 4) correlate with high H-density and, for S10, severely insufficient compute (only 11 minutes). Rebalancing the time budget would likely close much of S10’s gap.

Table 2: Final results on CAISc 2026 benchmarks (Lightning AI A100, 10h). S1–S4 are proven-optimal references. R_g^2 is the radius of gyration squared (compactness tiebreaker).

ID	Length	H%	Best known	Ours	Gap	R_g^2	Time
S1	20	55	9	9	0	—	ref
S2	24	54	9	9	0	—	ref
S3	25	36	8	8	0	—	ref
S4	36	50	14	14	0	—	ref
S5	48	50	23	23	0	8.1	48m
S6	50	44	21	19	2	10.4	48m
S7	60	67	36	35	1	10.5	67m
S8	64	66	42	38	4	13.1	79m
S9	85	68	53	51	2	14.8	171m
S10	100	54	50	46	4	18.7	11m
<i>Total contacts (S5–S10)</i>				212/225			94.2%

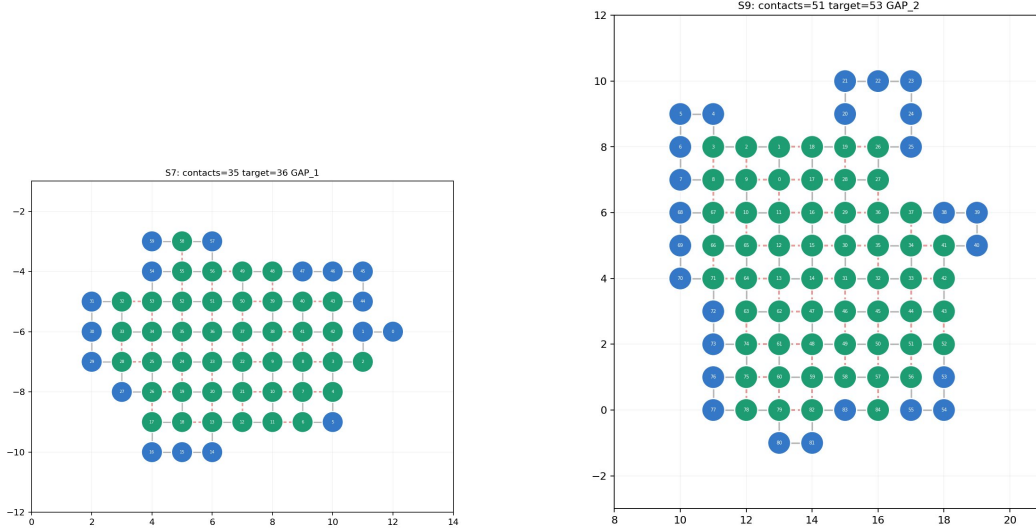


Figure 5: Best conformations for S7 (35 contacts, gap = 1) and S9 (51 contacts, gap = 2). Green = H, blue = P, dashed red = H–H contacts. Both show compact diamond-shaped hydrophobic cores characteristic of near-optimal folds.

4.6 Per-Sequence Analysis

S5 (48-mer) consistently reaches 23 contacts, matching the best-known value; its balanced 50% H-density gives a landscape where pivots locate the optimal core efficiently. S7 (60-mer) reaches 35 of 36, forming a dense diamond core (Figure 5); the missing contact likely needs a global rearrangement not found within 67 minutes. S9 (85-mer) achieves 51 of 53 in its 171-minute allocation, with the two missing contacts attributable to the P-rich tail (residues 64–85) not being optimally threaded through the core boundary. S10 (100-mer) has the largest gap but was also the most under-resourced, receiving only 11 minutes (Section 5).

4.7 Comparison with Prior Methods

Our solver matches the REMC results of Thachuk et al. [Thachuk et al.(2007)] on S5 and approaches the Wang–Landau results of Wüst and Landau [Wüst and Landau(2012)] on S9 (51 vs. 53), using a single-node 10-hour budget. We do not have precise compute figures from these prior works and therefore make no direct efficiency comparison; we note only that our results were obtained on a single GPU node without cluster resources. The remaining gap to Wang–Landau’s best-known values

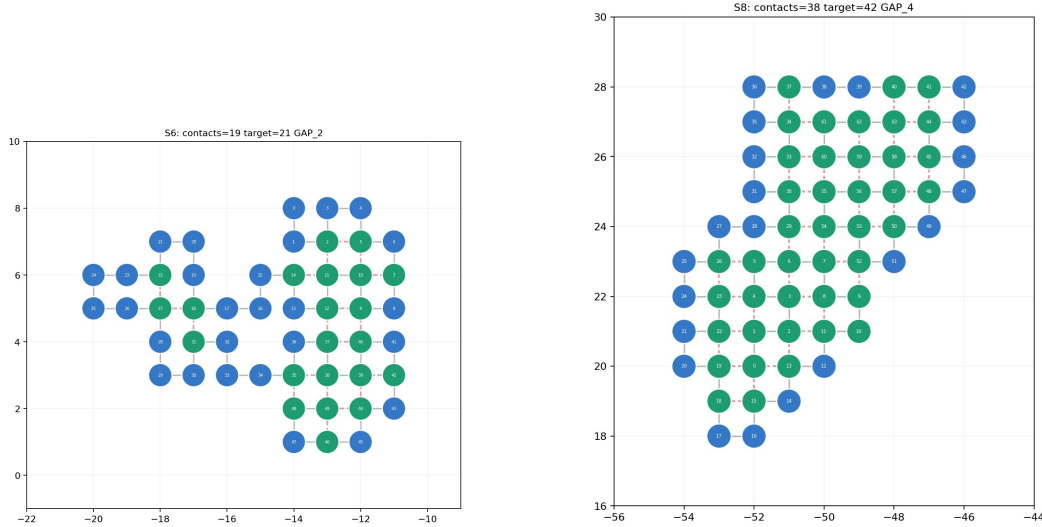


Figure 6: Best conformations for S6 (19 contacts, gap = 2) and S8 (38 contacts, gap = 4). S8’s high H-density (66%) creates a particularly rugged energy landscape with many competing near-optimal core arrangements.

Table 3: Massively parallel GPU annealing versus CPU REMC on the two hardest instances. Both were given comparable wall-clock budgets and use an identical move set; the GPU configuration runs $\sim 16,384$ concurrent chains but has no replica exchange. More chains did not translate into better folds.

Instance	Best known	GPU: parallel SA		CPU: REMC	
		Contacts	Gap	Contacts	Gap
S8 ($n=64$)	42	37	5	38	4
S10 ($n=100$)	50	45	5	46	4

is plausibly attributable to its flat-histogram property, which forces exploration of all energy levels—a capability that REMC with a fixed geometric temperature ladder does not provide.

The key advantage of our work relative to prior studies is not raw performance but *interpretability*: the ablation study (Table 1) provides direct quantitative evidence for move-operator design choices. While prior work discusses move operators qualitatively, we are not aware of a systematic operator-level isolation of contact-count contributions in this specific controlled form for the 2D HP model.

4.8 Does Massive Parallelism Substitute for Replica Exchange?

The ablation establishes that pivots dominate the move set. A natural follow-up: if pivots do most of the work, can raw parallelism replace replica exchange?

We ported the complete move set—pivot, corner, pull, end, with the same incremental- ΔE pivot—into a CUDA kernel where each thread anneals one independent chain to convergence using a per-thread occupancy grid. On $2 \times$ Tesla T4 this runs **16,384 concurrent chains** per launch over many rounds, two to three orders of magnitude more than the CPU configuration. The design is embarrassingly parallel: threads never communicate, so there is no replica exchange.

Despite roughly 10^3 times more concurrent chains, the GPU configuration was *worse* on both instances, losing one contact each (Table 3). Both GPU folds are valid self-avoiding walks with verifier-confirmed contact counts.

The interpretation is constructive. Independent chains, however many, each explore a single basin and cannot share progress: a chain that falls into a poor core topology early wastes its entire budget. Replica exchange couples chains through a temperature ladder, letting a good conformation found at

high temperature propagate down for low-temperature refinement. The ablation shows which *moves* matter; this experiment shows the *coupling between chains* contributes something extra chains do not replace. As design guidance: prefer fewer coupled replicas over many independent ones. This also tempers Section 4.3—faster pivots help, but throughput is not the binding constraint on quality.

We therefore report CPU REMC as our primary results (Table 2). Two caveats: warp divergence from variable-length pivot segments puts realised GPU throughput below peak, and this compares two design points rather than parallel SA and REMC in general.

5 AI-Driven Development

The development trajectory illustrates both the strengths and failure modes of LLM-led research:

Version 1 (no pivots). The initial solver used only corner flips, end flips, and simple pulls. Results were poor: S5 gap = 6, S8 gap = 13, S9 gap = 14. The AI overlooked the critical importance of large-scale structural moves despite their prominence in the cited literature.

The cascade-pull bug. In Version 2, the AI implemented a “cascade pull” requiring a lattice position adjacent to two bonded residues simultaneously. Mathematical analysis later revealed this is geometrically impossible on the 2D square lattice: two adjacent positions have disjoint 4-neighborhoods (sharing no common neighbor). The function’s precondition was never satisfied, meaning 25% of the move budget was silently wasted on dead code for an entire solver version.

Version 3. After removing the broken cascade pull and adding pivot rotations, crankshaft, and rebridging, S9 improved from gap = 14 to gap = 2. The grid-based pivot optimization further improved throughput by 1.4–1.5 $\times$.

Sequence hallucination. The AI initially used incorrect sequences for S6 and S10 from parametric memory rather than verifying against the competition source. Runtime verification against the live JSON caught this before submission.

Scheduling error. The time-budget loop terminated early for S10, allocating it only 11 minutes instead of its intended multi-hour share. Because S10 is the 100-mer (largest search space), this under-allocation is the primary cause of its gap-of-4 result; the error was in the AI-written scheduling logic, not the search algorithm itself.

These failure modes—geometric impossibility of a proposed algorithm, silent dead code, hallucinated benchmark data, and a compute-scheduling error—are characteristic of current LLM-driven development and underscore the necessity of runtime verification and careful auditing of AI-generated infrastructure code, not just AI-generated scientific claims.

5.1 Limitations and Future Work

Two directions could close the remaining gaps: Wang–Landau flat-histogram sampling [Wüst and Landau(2012)], which forces exploration of all energy levels rather than relying on thermal fluctuation, and adaptive temperature spacing driven by exchange-rate feedback. Porting the incremental- ΔE pivot (used in our GPU kernel, Section 4.8) into the CPU REMC path would remove the $O(n^2)$ recompute, though Section 4.8 suggests throughput is not the binding constraint.

6 Conclusion

We presented a six-operator REMC solver for 2D HP protein folding. Controlled ablation shows pivot rotations account for $\sim 80\%$ of the improvement over local-move-only search; a second controlled experiment shows that 16,384 concurrent independent GPU chains—identical move set—do not match REMC, indicating inter-chain coupling contributes what raw parallelism cannot. Both results argue that performance rests on move quality and replica coupling rather than throughput. On CAISc 2026 the solver matched S5 and achieved 212/225 contacts (94.2%). The documented AI failures—an impossible move operator, hallucinated sequences, missing pivots, and a scheduling error—evidence concrete LLM failure modes and the role of experimental validation in AI-driven science.

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Conference For AI Scientists 2026 - AI Involvement Checklist

Research Stage Assessment

1. Hypothesis development:

Answer: **[D]**

Iteration Effort: **[Medium]**

Explanation: The AI system selected the HP protein folding problem from five verifiable options by analyzing feasibility, search-space structure, and verification complexity. It reviewed the literature spanning Lesh (2003), Thachuk (2007), Wüst & Landau (2012), and Khandoker (2025), identifying the move-operator design space and REMC as the most viable framework. The human provided the conference call for papers and compute constraints.

2. Experimental design and implementation:

Answer: **[D]**

Iteration Effort: **[High]**

Explanation: The AI designed and implemented the solver in Python/Numba across three major versions: (v1) SA-only without pivots, (v2) REMC with pivots and a broken cascade pull, (v3) removal of the geometrically impossible cascade pull, addition of crankshaft and rebridging, and the grid-based pivot collision optimization. Over 100 substantive code iterations were performed, including debugging a silent-failure cascade-pull bug. The AI also designed and executed all four ablation experiments, and independently proposed, implemented (as a CUDA kernel), and evaluated the massively-parallel GPU comparison of Section 4.8, whose negative outcome contradicted its own prior expectation.

3. Analysis of data and interpretation of results:

Answer: **[D]**

Iteration Effort: **[Medium]**

Explanation: The AI analyzed checkpoint JSONs, identified the $O(n \cdot s)$ pivot bottleneck via complexity analysis, designed and ran ablation experiments, generated all figures, and interpreted results including the cascade-pull geometric impossibility proof. The human uploaded raw outputs and conformation images.

4. **Writing:**

Answer: [\[D\]](#)

Iteration Effort: [\[Medium\]](#)

Explanation: The AI wrote the complete paper including all sections, algorithm pseudocode, tables, figure captions, references, and both checklists. The human provided the $\text{\LaTeX}$ template.

AI System Documentation

5. **AI systems used:**

Description: A frontier large language model (LLM) served as the primary research agent across all stages of the research process: problem selection, literature review, algorithm design, Python/Numba implementation, iterative debugging, ablation experiment design, performance optimization, result analysis, and paper composition. The LLM operated in interactive conversational mode with a human supervisor who provided compute resources (Kaggle T4 GPUs, Lightning AI A100) and high-level direction. The solver itself is a classical REMC algorithm; no neural networks, reinforcement learning agents, or learned components were used in the search procedure.

6. **Observed AI Limitations:**

Description: Four significant failure modes were observed during development: (1) The AI implemented a cascade-pull move whose geometric precondition is impossible on the 2D square lattice—two bonded lattice positions share no common neighbor, so the precondition was never satisfied. This dead code silently consumed 25% of the move budget for an entire solver version. (2) The AI initially used incorrect benchmark sequences for S6 and S10 from parametric memory, producing spurious improvement claims. Runtime verification against the competition JSON caught this error. (3) The first solver version lacked pivot moves entirely, despite their prominence in the cited literature, producing results far below state of the art. The importance of pivots was only recognized after analyzing poor initial results. (4) An error in the AI-written time-budget scheduler terminated the loop early for S10, allocating it 11 minutes instead of its intended multi-hour share; this was infrastructure code rather than search logic, and was caught only when the reported per-sequence timings were audited against the intended allocation. Separately, the AI’s expectation that massive GPU parallelism would improve results proved incorrect (Section 4.8); we report this negative result rather than discarding it.

Conference For AI Scientists 2026 - Reproducibility and Responsibility Checklist

1. **Claims**

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [\[Yes\]](#)

Justification: The abstract states we matched S5 and reached 94.2% of contacts across S5–S10; Table 2 confirms this. The ablation claim is supported by Table 1 and Figure 1. The pivot speedup is measured in Figure 2.

2. **Limitations**

Question: Does the paper discuss the limitations of the work performed by the AI system(s)?

Answer: [\[Yes\]](#)

Justification: Section 5 discusses three specific AI failure modes. Section 5.1 discusses remaining gaps, S10 under-allocation, and high-H-density difficulty.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper include the full set of assumptions and a complete (or correct) proof?

Answer: [NA]

Justification: No new theorems are proved. NP-completeness is cited from prior work. The pivot collision correctness argument is informal.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper?

Answer: [Yes]

Justification: Full source code (Jupyter notebook) is submitted as supplementary material. All hyperparameters are specified in Section 3. Benchmark sequences are verified at runtime against the live competition JSON.

5. Open access to data and code

Question: Does the paper provide open access to the data and code?

Answer: [Yes]

Justification: The complete Jupyter notebook, all checkpoint JSONs, and submission coordinate files for S5–S10 are provided as supplementary material on OpenReview.

6. Experimental setting/details

Question: Does the paper specify all the training and test details necessary to understand the results?

Answer: [Yes]

Justification: Section 4.1 specifies the hardware platform (Lightning AI A100-40GB, 4 CPUs), wall-clock budget (10 hours), and software stack (Python 3, Numba JIT). Section 3 details all hyperparameters including temperature bounds, move probabilities, replica counts, and compute scaling rules.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly?

Answer: [No]

Justification: The benchmark results (Table 2) are from a single 10-hour run. The ablation study (Table 1) reports both best and mean over 10 independent seeds. Full variance analysis across multiple long runs was not feasible within the competition deadline.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources needed to reproduce them?

Answer: [Yes]

Justification: Lightning AI A100-40GB instance, 4 CPU cores, 10-hour wall clock. Per-sequence compute times are reported in Table 2. Ablation experiments specify step counts and seed counts.

9. Research Ethics

Question: Does the research conducted in the paper conform, in every respect, with ethical norms?

Answer: [Yes]

Justification: This work solves a well-known combinatorial optimization benchmark. No human subjects, sensitive data, or dual-use concerns are involved.

10. Broader impacts

Question: Does the paper discuss both potential positive and negative societal impacts of the work performed?

Answer: [NA]

Justification: The HP model is a simplified abstraction used for algorithmic research. This work does not produce real protein structures and has no direct societal impact, positive or negative.